

# Foundations of Robotics

Study the official course objectives in the source syllabus.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 2331 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 2331.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item             | Official information             |
|------------------|----------------------------------|
| Programme        | Diploma Programme                |
| Course code      | 2331                             |
| Course title     | Foundations of Robotics          |
| Semester         | II                               |
| Course category  | As specified in the official PDF |
| Periods per week | As specified in the official PDF |

| Item                 | Official information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Periods per semester | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Credits              | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Prerequisite         | Topic Course code Course Title Semester Basic knowledge in Physics 2002B Engineering Physics for Applied Electrical Technology and Computing I Basic knowledge in Chemistry 2003A Chemistry for Engineering Practices I Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Select and use sensors in various robotic application contexts Apply CO2 Select and use actuators in various robotic application contexts Apply CO3 Explain the broad concepts of robotics and its applications Understand CO4 Develop basic kind of autonomous robot Apply CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES |
| Assessment           | Use the official assessment section reproduced below                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Modules

4 official module sections detected.

## Topic entries

8 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

## Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

## LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

- Study the official course objectives in the source syllabus.

### Course outcomes

| CO  | Official outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Study level           |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 3 2 - 3 2 - - - - -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | See official syllabus |
| CO2 | 2 3 3 2 2 - - - - -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | See official syllabus |
| CO3 | 2 2 - 3 3 3 - - - - 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | See official syllabus |
| CO4 | 3 3 3 2 3 3 - - - - 3 Total 10 10 6 10 10 6 - - - - 6 Correlation Level 2.5 2.5 3 2.5 2.5 3 - - - - 3 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-” Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 Diploma Curriculum Revision 2026 2331 Page #1 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Hours Module Title Major Topics L T | See official syllabus |

#### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3 2 - 3 2 - - - - -    |
| CO2 | CO2 2 3 3 2 2 - - - - -    |
| CO3 | CO3 2 2 - 3 3 3 - - - - 3  |
| CO4 | CO4 3 3 3 2 3 3 - - - - 3  |

#### COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

#### Module coverage

| Module   | Official heading                                                                                   | Topic entries | Hours        |
|----------|----------------------------------------------------------------------------------------------------|---------------|--------------|
| Module 1 | Sensors in Robotics Basics of Sensors, Internal Sensors 15 0                                       | 1             | 15           |
| Module 2 | Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0               | 1             | 17           |
| Module 3 | Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0                    | 1             | 13           |
| Module 4 | Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 | 5             | As specified |

#### MODULE 1

## Sensors in Robotics Basics of Sensors, Internal Sensors 15 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

15

CO-linked

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Sensors in Robotics Basics of Sensors, Internal Sensors 15 0
- Official duration: 15
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

## Sensors in Robotics Basics of Sensors, Internal Sensors 15 0

Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 is an official topic in Module 1 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, sensors in robotics basics of sensors, internal sensors 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------|
| Definition/identity | What Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1-സെൻസറുകൾ Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 സെൻസറുകളുടെ നിർവ്വചനം/അർത്ഥം, സെൻസറുകളുടെ പ്രയോഗം, steps, application സെൻസറുകളുടെ പ്രാധാന്യം. Technical terms English-ൽ സെൻസറുകൾ സെൻസറുകളുടെ പ്രാധാന്യം.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

|                           |    |   |           |
|---------------------------|----|---|-----------|
| Sensors in Robotics       |    |   | Basics of |
| Sensors, Internal Sensors | 15 | 0 |           |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

**17**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## **Key facts**

- Official module title: Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0
- Official duration: 17
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.



## Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0

Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 is an official topic in Module 2 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, actuators, drives and advanced robotics applications building blocks of a robot 17 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-റോബോട്ടിക്സ് ആക്ടുേറ്ററുകൾ, ഡ്രൈവുകൾ, ഉപയോഗ റോബോട്ടിക്സ് ആപ്ലിക്കേഷനുകൾ  
Building Blocks of a Robot 17 0 റോബോട്ടിക്സ് ആക്ടുേറ്ററുകൾ, ഡ്രൈവുകൾ, ഉപയോഗ റോബോട്ടിക്സ് ആപ്ലിക്കേഷനുകൾ  
definition/meaning റോബോട്ടിക്സ് ആക്ടുേറ്ററുകൾ, ഡ്രൈവുകൾ, ഉപയോഗ റോബോട്ടിക്സ് ആപ്ലിക്കേഷനുകൾ, steps, application  
റോബോട്ടിക്സ് ആക്ടുേറ്ററുകൾ, ഡ്രൈവുകൾ, ഉപയോഗ റോബോട്ടിക്സ് ആപ്ലിക്കേഷനുകൾ. Technical terms English-റോബോട്ടിക്സ് ആക്ടുേറ്ററുകൾ, ഡ്രൈവുകൾ, ഉപയോഗ റോബോട്ടിക്സ് ആപ്ലിക്കേഷനുകൾ.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Actuators, Drives and Advanced Robotics Applications  
Building Blocks of a Robot 17 0

Building

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

13

official hours

CO-linked

see outcomes

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### **Key facts**

- Official module title: Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0
- Official duration: 13
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

## Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0

Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 is an official topic in Module 3 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, fundamentals of robotics & applications definitions of robots and robotics 13 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3- ഫണ്ടമെന്റല ഓഫ് റോബട്ടിക്സ് & അപ്ലിക്കേഷൻസ്  
Definitions of Robots and Robotics 13 0 ഡിഫിനിഷൻ/മീനിംഗ്  
പ്രദാനം ചെയ്യുന്നു. റോബട്ടിക്സ് റോബട്ടിക്സ്, steps, application റോബട്ടിക്സ്  
പ്രദാനം ചെയ്യുന്നു. Technical terms English-ൽ റോബട്ടിക്സ് പ്രദാനം ചെയ്യുന്നു.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

|                                         |    |   |
|-----------------------------------------|----|---|
| Fundamentals of Robotics & Applications |    |   |
| Definitions of Robots and Robotics      | 13 | 0 |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 4

# Autonomous Robots - Line Follower and Obstacle Avider Basic Principle of Line Follower Robot 15 0

This module is presented from the official syllabus scope for Foundations of Robotics.  
Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### **Key facts**

- Official module title: Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0

Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, autonomous robots – line follower and obstacle avoider basic principle of line follower robot 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-രണ്ടാം Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 രണ്ടാംരണ്ടാംരണ്ടാംരണ്ടാം രണ്ടാംരണ്ടാം definition/meaning രണ്ടാംരണ്ടാംരണ്ടാംരണ്ടാം. രണ്ടാംരണ്ടാം രണ്ടാംരണ്ടാം രണ്ടാംരണ്ടാം, steps, application രണ്ടാംരണ്ടാം രണ്ടാംരണ്ടാംരണ്ടാം രണ്ടാംരണ്ടാം. Technical terms English-രണ്ടാം രണ്ടാം രണ്ടാംരണ്ടാംരണ്ടാം.



## Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Detailed Syllabus means in the official course context                       |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Detailed Syllabus തിരഞ്ഞെടുത്ത വിഷയത്തിന്റെ definition/meaning നൽകുക. വിഷയത്തിന്റെ ഏതെങ്കിലും steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

## Mode of

Mode of is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Mode of means in the official course context                                 |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-Mode of  $\frac{a+b}{2}$  definition/meaning  $\frac{a+b}{2}$ .  $\frac{a+b}{2}$  steps, application  $\frac{a+b}{2}$ . Technical terms English-Mode of  $\frac{a+b}{2}$ .

## Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Mapped Instructional means in the official course context                    |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്ന Mapped Instructional ഏകീകൃതതയുള്ള പാഠ്യപുസ്തകം definition/meaning ഉൾക്കൊള്ളുന്നു. പാഠ്യപുസ്തകം പാഠ്യപുസ്തകം, steps, application ഉൾക്കൊള്ളുന്നു. Technical terms English-ൽ പാഠ്യപുസ്തകം ഉൾക്കൊള്ളുന്നു.

## Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Definition/identity | What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context |
| Study lens          | Principle → components or stages → result → application                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-Subtopic Student Learning Outcome delivery Taxonomy Level definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

|                                                        |                          |   |       |
|--------------------------------------------------------|--------------------------|---|-------|
| Autonomous Robots – Line Follower and Obstacle Avoider |                          |   | Basic |
| Principle of Line Follower Robot                       | 15                       | 0 |       |
| Detailed Syllabus                                      |                          |   |       |
| Mode of                                                |                          |   |       |
| Mapped                                                 | Instructional            |   |       |
| Subtopic                                               | Student Learning Outcome |   |       |
| delivery                                               | Taxonomy Level           |   |       |
| C0                                                     | Hours                    |   |       |
| (L/T)                                                  |                          |   |       |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## RAPID REFERENCE

## Concept and formula bank

### Answer structure



Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

## Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

### RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Prepare notes on internal sensors such as position, velocity sensors, and accelerometers with applications. 6
- Study external sensors including proximity, force, and torque sensors and identify their real-time uses in robotics. 6
- Create a table mapping different sensing variables (temperature, light, humidity, sound) with suitable sensors. 6
- Prepare chart for all sensors and take a seminar each sensors 6 Module II
- Study different types of electric motors (DC, servo, stepper) used as actuators in robotic systems. 6
- Apply speed and direction control techniques of motors and selection criteria for actuators in robots 6
- Case study on selection of sensors and actuators for self-driving cars and maze-solving robots (advanced content). 6 Module III
- Study and summarize the definitions, need, types, and characteristics of robots using textbook and NPTEL resources. 6
- Prepare a comparative chart on advantages and limitations of robots in different application areas such as medical, defense,
- and domestic sectors.
- Explore real-world industrial robot applications (material handling, welding, painting) through videos and case studies. 6
- Prepare report on case studies based robot applications 6 Module IV
- Study the basic principle of line follower robots and color sensing using online simulations and reference videos. 6
- Identify different sensors used in line follower and obstacle avoiding robots and explain their working principles. 6
- Analyze the block diagram of a line follower robot and explain signal flow from sensor to actuator 6
- Total Self-Learning hours 90 Learning Resources TEXT BOOK
- Title Author Publication
- Robotics for Engineers Yoram Koren McGraw-Hill Education

## EXAMINATION PREPARATION

# Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

|               |               |                 |                 |               |    |
|---------------|---------------|-----------------|-----------------|---------------|----|
| – Theory      |               |                 |                 |               |    |
|               |               | CA1             |                 | CA2           |    |
| CA3           | CA4           | CA5             | CA6             | ESE           |    |
| Mode          | Attendance    | Self-learning   | Self-learning   | Written       |    |
| Self-learning | Self-learning | Series          | Series          | activities    |    |
| activities    | activities    | activities      | activities      | Exam          |    |
| (Module 3)    | (Module 4)    | Exam (Module 1) | Exam (Module 2) | Exam (theory) |    |
| Duration      |               |                 |                 |               |    |
| 2 hours       | 2 hours       | 2.5 hours       |                 |               |    |
| Exam mark     |               |                 |                 |               |    |
| 15            | 15            | 50              | 15              | 50            | 15 |
| Converted to  |               |                 |                 |               |    |
| 20            | 20            |                 |                 |               |    |
| Marks         | 5             |                 |                 |               |    |
| 20            | 60            |                 |                 |               |    |
| Total         |               |                 |                 |               |    |
| 40            |               |                 |                 |               |    |
| Tentative     | End of        |                 |                 |               |    |
| End of        |               |                 |                 |               |    |
| 11th week     | By 14th week  | By 4th week     | By 7th week     | By            |    |
| schedule      | semester      | 8th week        | 15th week       |               |    |
| semester      |               |                 |                 |               |    |

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Sensors in Robotics Basics of Sensors, Internal Sensors 15 0. (3 marks)

**Answer:** Begin with the standard definition or identity of *Sensors in Robotics Basics of Sensors, Internal Sensors 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 2

**Define or explain Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

**Define or explain Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 4

**Define or explain Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Detailed Syllabus. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Detailed Syllabus*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Mode of. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Mode of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Mapped Instructional. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Mapped Instructional*. State the governing principle, list the key components or stages, and close with one relevant

application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **Sensors in Robotics Basics of Sensors, Internal Sensors 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

### REFERENCE VOCABULARY

# Glossary

## Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

- URL Modules Covered
- <https://nptel.ac.in> NPTEL Robotics Courses
- <https://create.arduino.cc/projecthub> Arduino Project

## IN-PAGE SEARCH

### Search this lesson

Prepared from the supplied official SITTTT syllabus PDF for Course 2331. Verify institutional updates against the current official curriculum.